

# Finn Kliewer

<https://www.finnkliewer.com> | <https://github.com/FinnKliewer> | <https://www.linkedin.com/in/finnkliewer>  
finn.kliewer@gmail.com | 201.747.2660

## PROFESSIONAL SUMMARY

Machine learning-focused software engineer specializing in production LLM systems, backend APIs, and cloud-native infrastructure. Experience building and scaling GenAI applications, RAG pipelines, and distributed services supporting millions of users.

## EDUCATION

**Cornell University — SC Johnson College of Business**

May 2024 – Jan 2026

Master of Science in Business Analytics

**Rutgers University — School of Arts and Sciences**

September 2021 – May 2024

Bachelor of Science in Computer Science, Minor in Mathematics

## KEY SKILLS

**Languages:** Python, C#, Java, C/C++, PHP, JavaScript, SQL

**ML & GenAI:** PyTorch, Scikit-learn, XGBoost, FAISS, Embeddings, RAG pipelines, LLM orchestration, LlamaIndex

**Backend & APIs:** FastAPI, GraphQL, REST APIs, Microservices

**Cloud & Infrastructure:** AWS (S3), Kubernetes, Docker, CI/CD (GitLab)

**Data Systems:** Redis, RabbitMQ, ETL pipelines, Splunk

## PROFESSIONAL EXPERIENCE

**Software Engineer II — Paycom**

August 2024 – Present

Built and scaled production backend and AI-driven systems at a \$12B HR technology company serving 36,000+ clients.

- Built and scaled backend infrastructure for IWant, Paycom's flagship LLM-powered assistant serving 3M+ monthly queries, by developing Python GraphQL APIs, reusable services and permissions frameworks, and LlamaIndex-based orchestration, improving response latency by 20% and reducing feature integration time.
- Led a team of 6 interns to migrate a legacy PHP service to scalable C# microservices using RabbitMQ, Docker, and Kubernetes, improving job runtime by 50%, reducing integration failures, and saving 600+ engineering hours annually; awarded Best Intern Project.
- Helped build a high-throughput AWS S3-based data pipeline for external credit reporting, enabling reliable access to 80M+ employee records and supporting sub-second API response times (<500ms)
- Maintained CI/CD pipelines, monitored systems with Splunk, and set up alerts, reducing deployment issues and improving production observability.

**Summer Analyst — Tiger Advisory**

Summer 2024

Developed production data and ML-backed backend systems in a consulting environment.

- Built and deployed a FastAPI-based NLP service to extract and structure applicant data, reducing manual processing time by 30+ minutes per record.
- Developed production CI/CD pipelines with unit testing, improving reliability and ensuring consistent data ingestion.
- Designed backend APIs integrating CRM and cloud storage systems, improving workflow automation and data accessibility.

## TECHNICAL PROJECTS

**Rutgers Bus AI Assistant**

<https://github.com/FinnKliewer/ru-bus-ai>

LLM-powered campus transit assistant with real-time data and interactive UI.

- Developed a Generative AI (GenAI) assistant for Rutgers students using Python, FastAPI, Redis, and PostgreSQL, enabling natural language interaction with real-time transit data.
- Designed a retrieval-augmented generation (RAG) pipeline using FAISS vector embeddings, enabling students to query live transit data using slang, landmarks, and nicknames, significantly improving contextual understanding and response accuracy.
- Implemented LLM orchestration and tool-calling workflows to generate both conversational responses and structured UI widgets for trip planning.

**EarlyTrace.ai**

<https://github.com/FinnKliewer/early-trace>

Python-based NLP system for early detection of dementia from text samples; runner-up at the 2025 Rutgers Hackathon.

- Trained an XGBoost classifier on 5,000 text embeddings with 5-fold cross-validation, achieving an F1 score of 0.95.
- Applied LIME for feature attribution and interpretability, improving model transparency.

**GLSL Mandelbrot Fractal**

<https://github.com/FinnKliewer/pyopengl-mandelbrot>

High-performance real-time fractal visualization tool.

- Built a real-time Mandelbrot Set explorer in Python using GLSL and PyOpenGL, implementing FP64 double-precision arithmetic, escape-time iteration, and smooth coloring with interactive zoom.